

Improving Maternal Health Risk Classification Using Machine Learning

Baseline Group Presentation
Eric Wu, John Bang, Nazia
Haque



Research Question & Motivation

Research Question:

- How can machine learning methods improve the accuracy and efficiency of maternal health risk classification during pregnancy?

Why It's Interesting:

- ~810 maternal and 6,700 newborn deaths daily (WHO).
- Many maternal deaths are preventable.
- Early, accurate risk prediction can save lives, especially in under-resourced regions.
- Deep learning models can run quickly on local devices, no GPU needed.



Dataset Overview

- **Source:** UCI Machine Learning Repository
- **Origin:** IoT monitoring systems in rural Bangladesh

Size:

- 1017 Samples, 7 features

Variables:

- Features: Age, SystolicBP, DiastolicBP, Blood Sugar, Body Temp, Heart Rate
- Target: RiskLevel (Low, Mid, High)

Missing Data:

- Lacks previous pregnancies, complications, miscarriages

Data Summary

Summary statistics for each variable

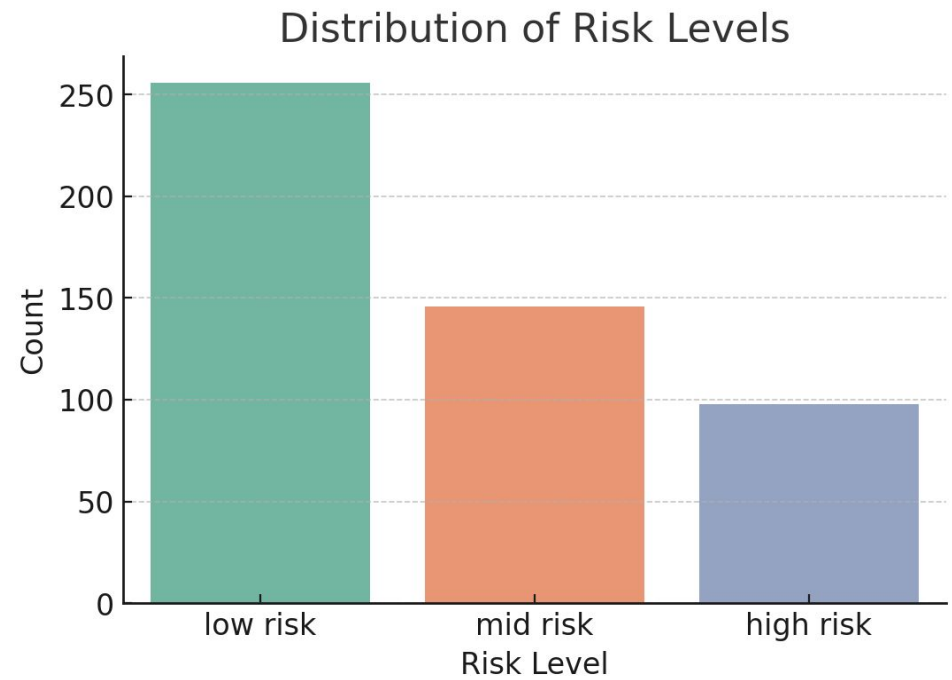
Distribution plots: Age, BP, BS, etc.

RiskLevel class balance: % Low, Mid, High

Summary Statistics of Maternal Health Features

	Age	SystolicBP	DiastolicBP	BS	BodyTemp	HeartRate
Mean	29.19	110.55	75.42	8.35	98.69	73.95
Std	13.77	17.87	13.75	2.83	1.41	8.16
Min	10.0	70.0	49.0	6.0	98.0	7.0
25%	19.0	90.0	65.0	6.9	98.0	70.0
50%	25.0	120.0	80.0	7.5	98.0	76.0
75%	35.0	120.0	86.0	7.9	98.0	80.0
Max	70.0	160.0	100.0	19.0	103.0	90.0

Risk Level Distribution



Algorithms



Baseline Model: Logistic
Regression

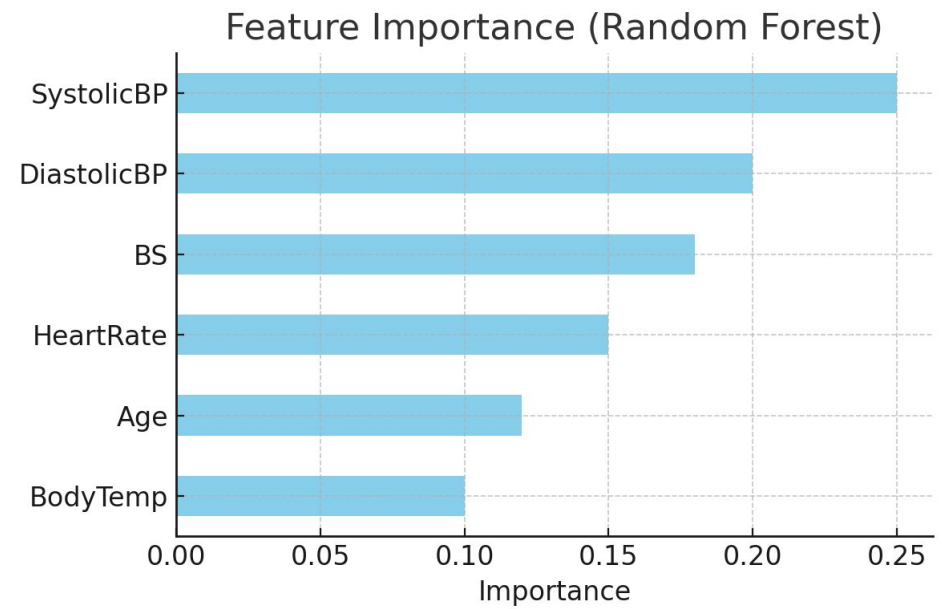


Random Forest: Ensemble,
handles feature importance
well



Artificial Neural Network:
Models nonlinear patterns,
lightweight for local use

Random Forest



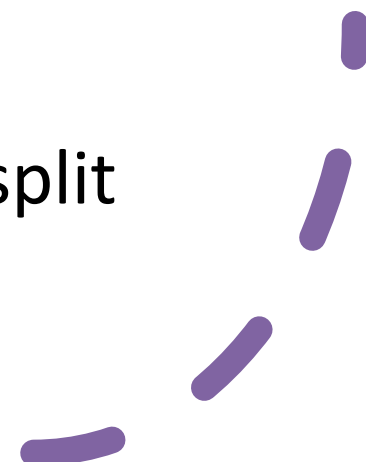
A large red circle on the left side of the slide, partially cut off by the edge.

Evaluation Plan

Metrics:

- Training, validation, and test accuracy
- F1 Score
- Confusion Matrix (class accuracy)

Data splits for model:

- Train-validation-test split (60/20/20)
- 
- A decorative purple dashed line in the bottom right corner of the slide.

Preliminary Results

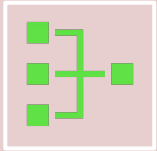


Initial performance of models:

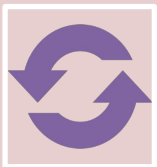
Logistic regression: TA: 0.6494, VA: 0.5889

Random Forest: TA: 0.9241, VA: 0.5441

ANN: TA: 0.7778, VA: 0.7260



ANN shows promising accuracy of ~70% for base model



Future work: refine models and tune hyperparameters

References

